NexaQuant

Evidence-First Quant Trading System

Gold (XAUUSD) + Bitcoin (BTCUSD) · Smart Money Concepts + Regime Gate + AI

TECHNICALS + SMC

structure · FVG · order blocks
liquidity · premium/discount

REGIME + RISK

trend/range/volatile gate
ATR stops · vol-target sizing

AI + FUNDAMENTALS

meta-labeling · MTF features
macro bias · ensemble-ready

Architecture & Evidence Deck

Charts in this deck are generated live from the real 2y gold dataset and the project's own backtests.

Executive Summary

Prove the edge, control drawdown, gate capital behind validation

The Thesis

Profit = edge x payoff x risk control — NOT win-rate or model hype. We capture a small, real, cost-robust edge (trend-continuation incl. SMC structure/FVG), cut drawdown hard with a regime gate, and let AI sharpen selection/sizing as data & features grow. Capital scales only behind a hard validation gate.

Regime-gated trend (H1)
Sharpe 3.6
out-of-sample, net of cost

Drawdown vs buy&hold
-33%
254 − > 172 (H1 OOS)

FVG + Structure (H4)
Sharpe 1.9
robust IN & OUT of sample

AI meta-label today
AUC 0.51
premature — needs data+features

VALIDATED (out-of-sample, net of cost)

+ Trend-continuation edge (EMA + FVG + Break of Structure)
+ Regime gate lifts Sharpe & cuts drawdown ~33%
+ Edge survives realistic spread/slippage
+ Multi-timeframe + fundamental feature pipeline wired & verified

NOT YET PROVEN (honest gaps)

- Much of return is GOLD BETA (2023-25 bull) — one regime only
- Mean-reversion SMC (discount, sweep-fade) lost in trend
- AI flat until M5/M15 + BTC + fundamentals add data/features
- No live/paper validation yet — do NOT deploy capital

The bet: invest in DATA + FEATURES + the VALIDATION GATE — that, not one magic model, is the asset.

System Architecture

Six layers — nothing reaches live capital without passing the gate

1 · DATA ENGINE
- MT5: XAUUSD + BTCUSD
- W1·D1·H4·H1 (analysis)
- M15·M5 (execution)*
- fundamentals (free)

2 · FEATURE & SMC
- Structure (BOS/CHoCH)
- Fair Value Gaps
- Order Blocks
- Liquidity sweeps
- Premium/Discount

3 · REGIME GATE
- trend / range / volatile
- ADX + vol ratio

PILLARS ON/OFF:
- trend->continuation
- range->mean-revert

4 · AI / SIGNAL
- MTF top-down features
- + fundamentals
- meta-label P(win)
- ensemble-ready

5 · RISK FORTRESS
- ATR stops & targets
- vol-target / Kelly-frac
- daily loss limit
- circuit breaker

6 · EXECUTION
- paper -> live ramp
- MT5 order routing
- slippage/latency model
- performance tracking

VALIDATION GAUNTLET (the real moat — between Signal and Execution)

Temporal firewall (embargo) → **Walk-forward + Purged CV** → **Regime-stratified backtest** → **30-day PAPER trading** → **Brier < 0.25 calibration** → **Go / No-Go certification**

Cross-cutting: config-driven (zero hardcoding) · health monitor & kill-switch · drift detection & auto-retrain · experience memory

** M15/M5 + BTCUSD not yet pulled (data/pull_mt5.py). Today: Data + SMC + Regime + Signal + Backtest run live.*

Data & Multi-Timeframe Stack

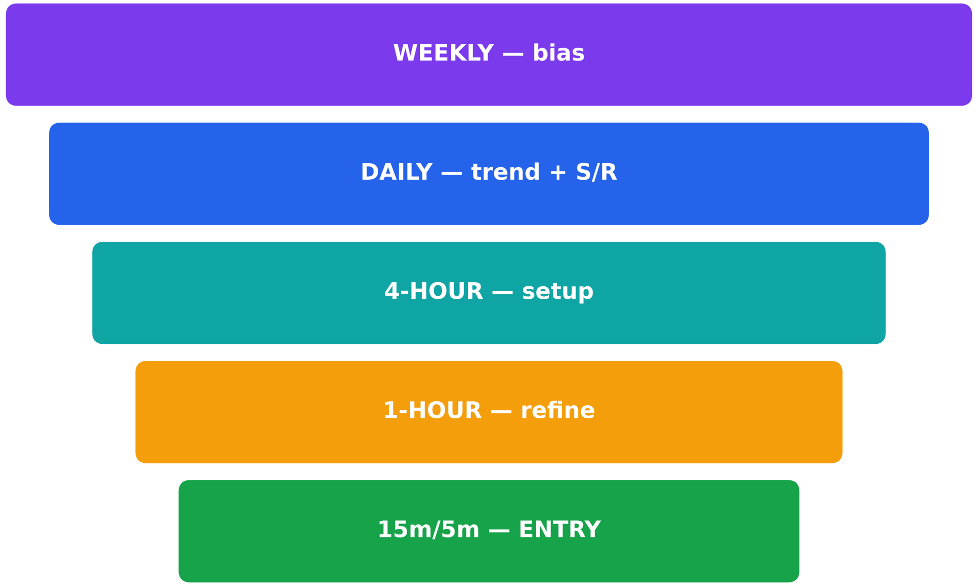
Analyse high, execute low — higher TFs become model features too

Instruments & timeframes

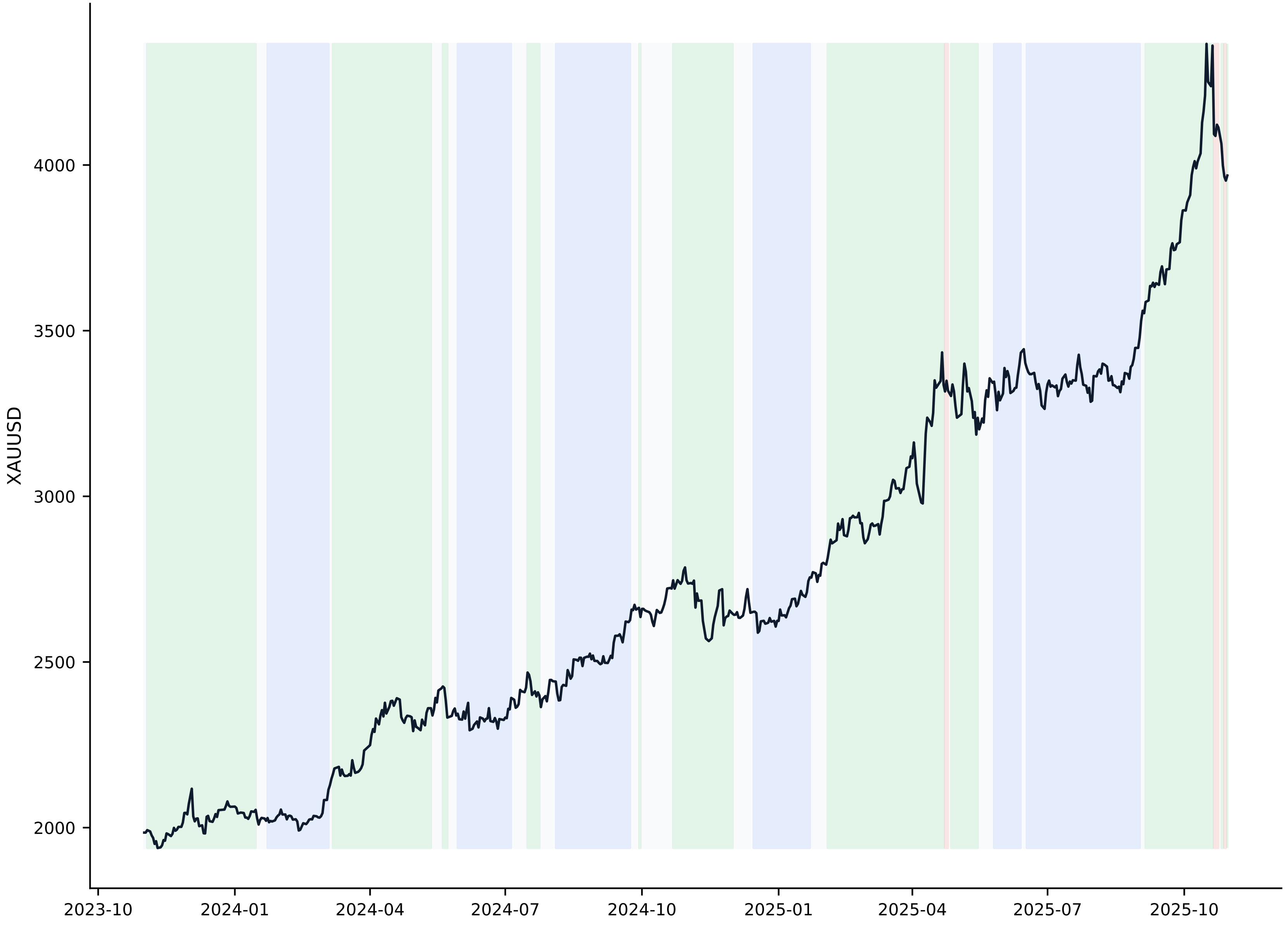
symbol	analysis	execution
XAUUSD (gold)	W1 D1 H4 H1	M15 M5
BTCUSD	W1 D1 H4 H1	M15 M5

Fundamentals (FREE): FRED real yields · DXY · CFTC COT · economic calendar

Top-down funnel (also the model's MTF features)



Gold (D1) coloured by detected regime — green=trend blue=range red=volatile



SMC Strategy Stack

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Gold + BTC Alpha

Read institutional footprints — the regime decides which to trust

The 5 pillars (with measured edge)

1 · Fair Value Gap — EDGE +
3-candle imbalance; price revisits

2 · Market Structure — EDGE +
BOS / CHoCH = trend & reversal

3 · Order Blocks — context
last opposite candle = demand/supply

4 · Liquidity Sweeps — EDGE - in trend
stop-hunt beyond swing then reclaim

5 · Premium / Discount — EDGE - in trend
25/50/75% of swing range

Regime gate decides

TRENDING
ON: Structure + FVG (continuation) — GO WITH the move

RANGING
ON: Discount-buy + sweep (mean-revert) — FADE extremes

VOLATILE
size DOWN or STAND ASIDE

KEY INSIGHT FROM OUR DATA

Continuation pillars (Structure + FVG) had POSITIVE edge on trending gold; mean-reversion pillars (deep-discount, sweep-fade) had NEGATIVE edge. Stacking all 5 ('A+') LOST money — the regime gate, not more indicators, is what turns SMC into profit.

Trading Playbook (the Strategy)

NexaQuant
Gold + BTC Alpha

Entries give a small edge — STOPS, TRAILING & MOMENTUM-RIDING make the profit

BIAS — Weekly/Daily trend + macro (yields↓, USD↓ = gold long)

REGIME GATE — trade continuation only in TREND; stand aside if volatile

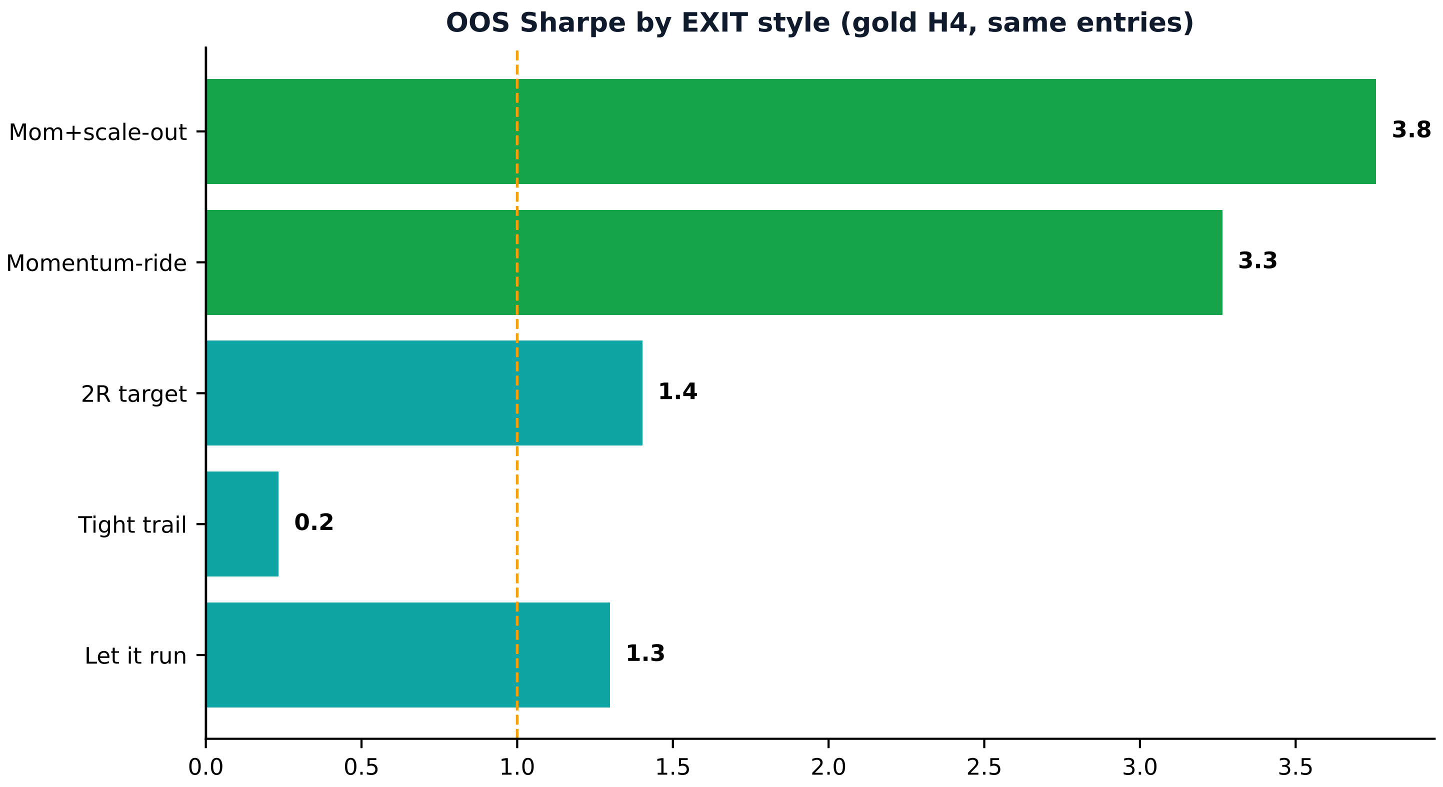
ENTRY — EMA20>EMA50 + bullish structure / FVG (execution TF)

STOP-LOSS — hard 2×ATR stop — NON-NEGOTIABLE, caps every loss

BIGGER PROFIT — MOMENTUM-RIDE: hold while close>EMA20, exit when it fades

SCALE-OUT — bank ~40% at +1.5R, move to breakeven, let the rest run

SIZING — volatility-targeted, fractional-Kelly capped



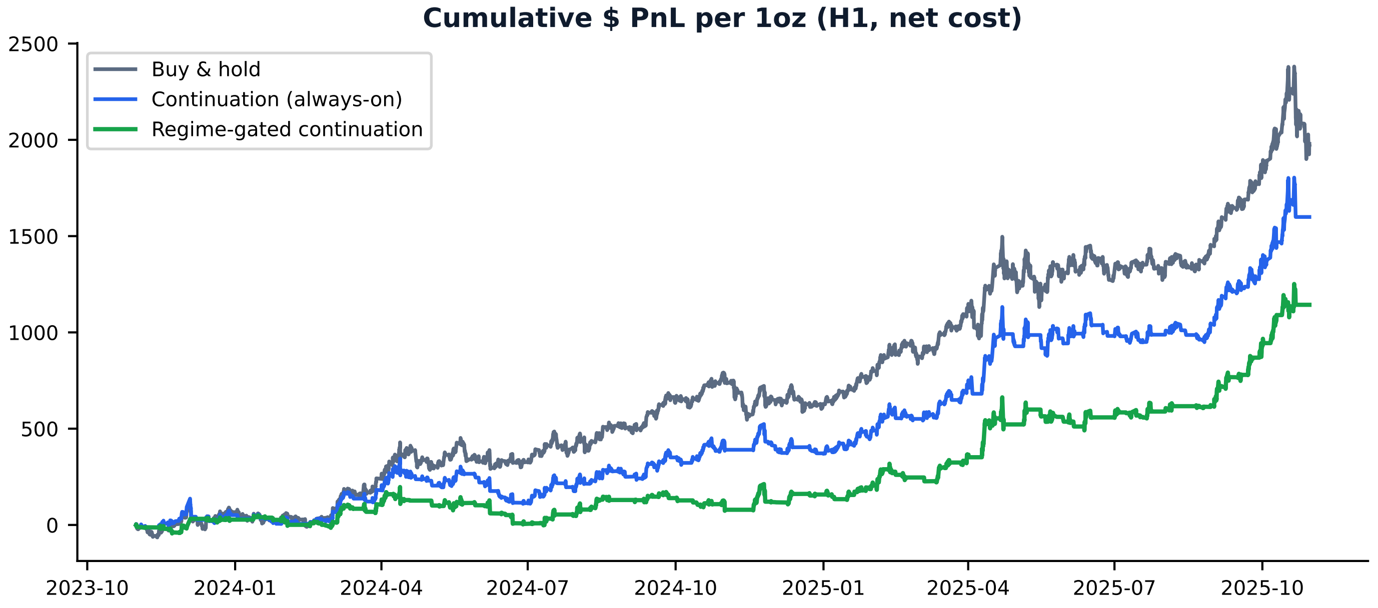
WHY MOMENTUM-RIDE

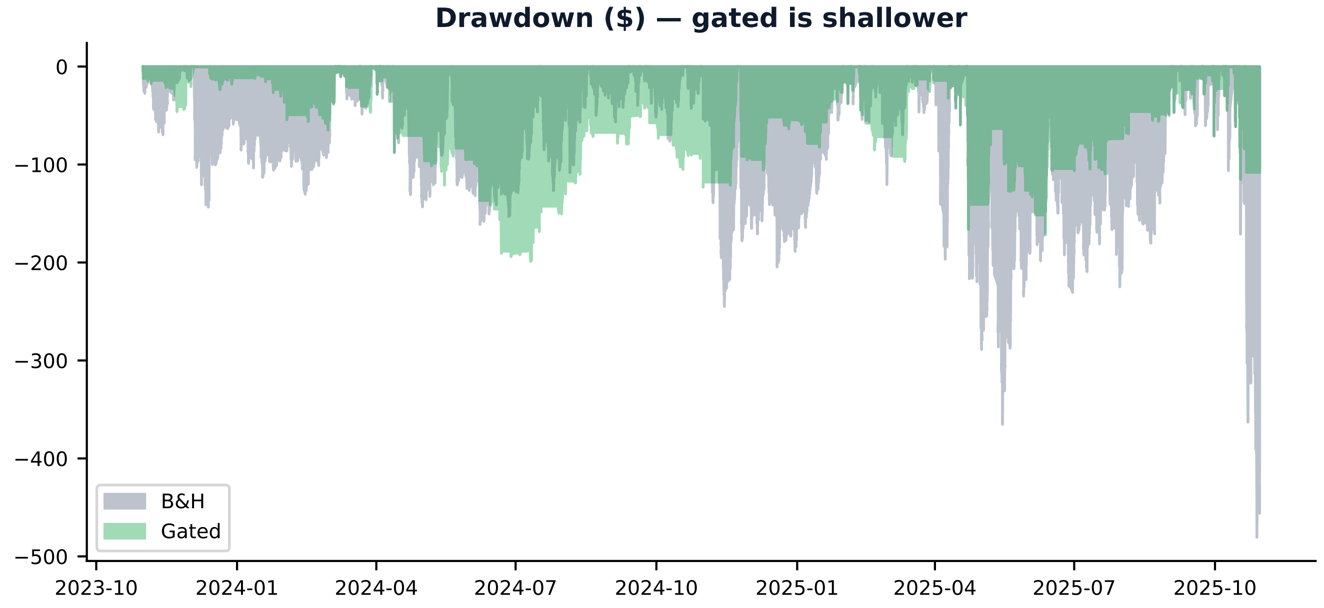
- Stop-loss is mandatory — caps every loser
- Tight trailing turns the edge NEGATIVE (chokes winners)
- Riding momentum captures big R-multiples (up to 8.7R) while cutting drawdown ~40% vs 'let it run forever'
- Scale-out lifts win-rate to ~64% and smooths the curve

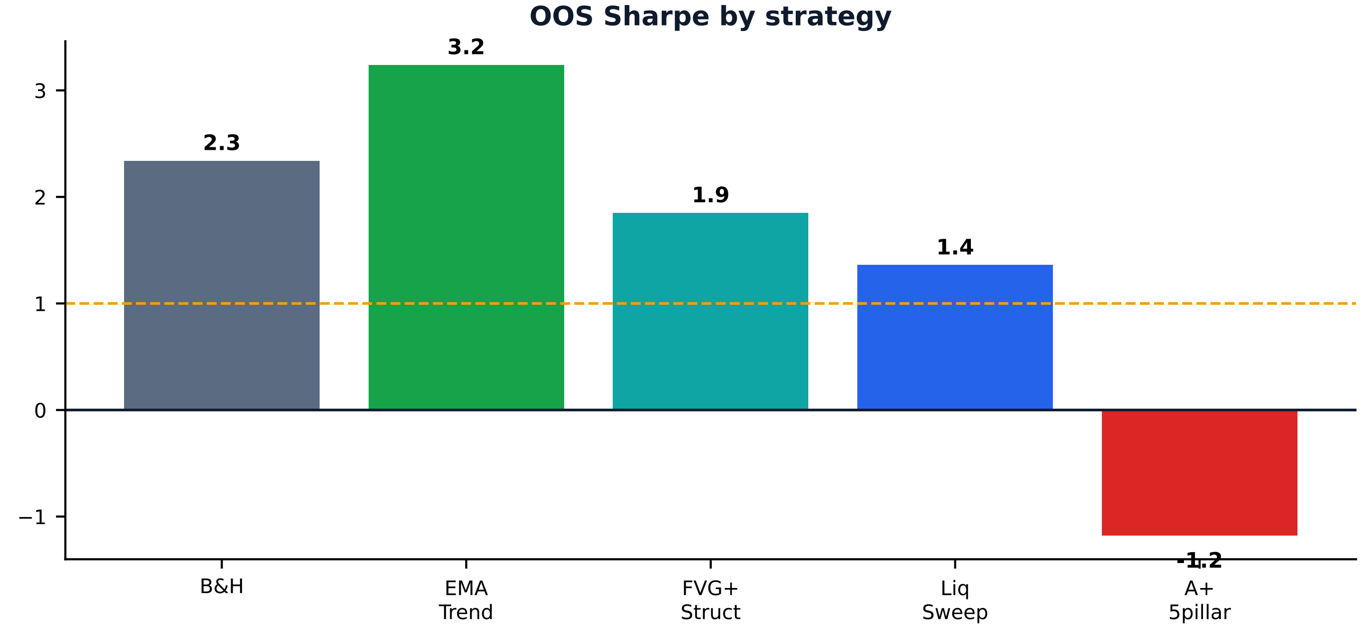
Evidence Dashboard

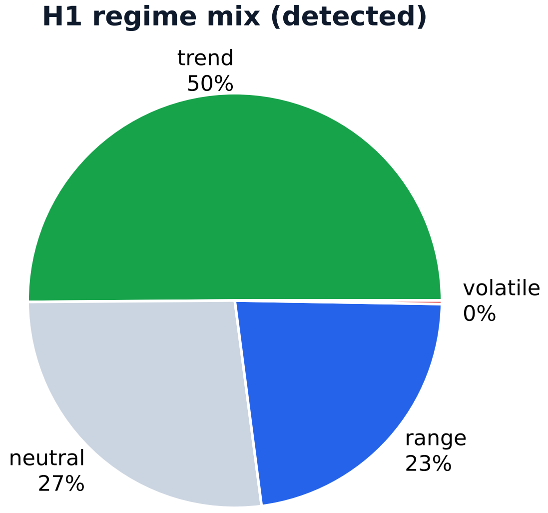
NexaQuant
Gold + BTC Alpha

Computed live from the real 2y gold data + project backtests





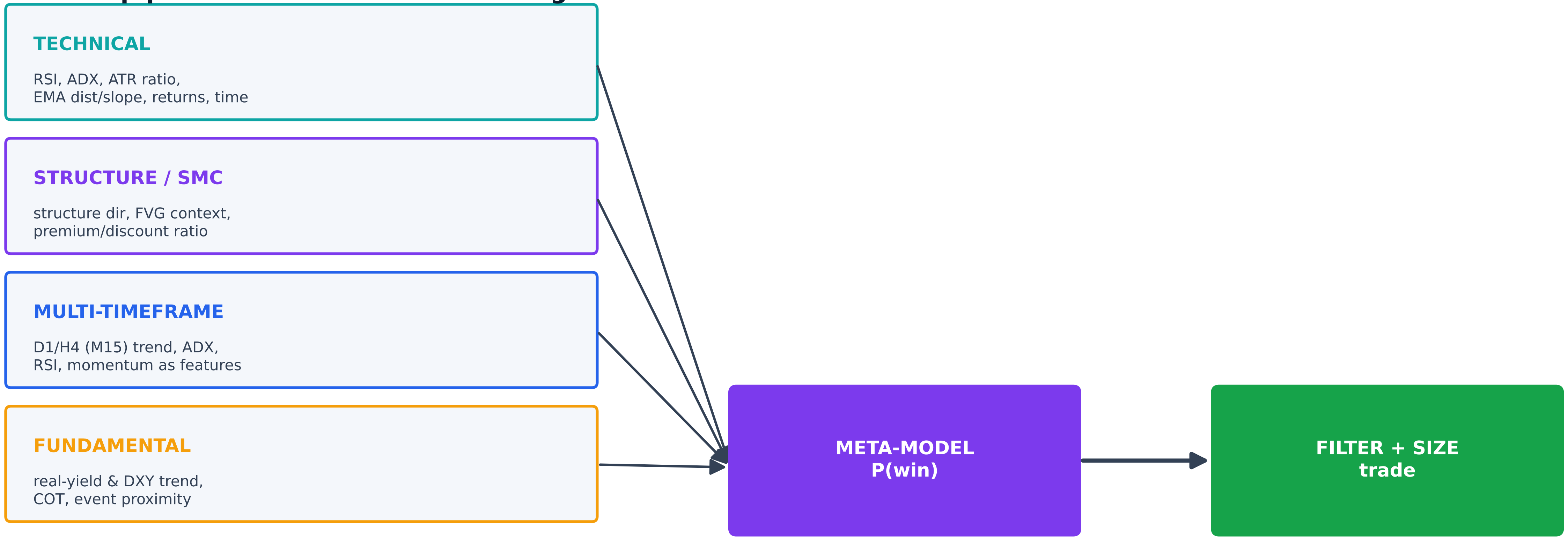




AI / Model Layer

Models SIZE & SELECT a known edge — they do not invent one

Feature pipeline -> meta-model -> sizing



Today: AUC 0.51 (coin-flip) — 154 entries, technical-only. The plumbing is verified (28/28 features active with MTF+fundamentals wired); it gains skill as data/features grow.

Model staging (complexity rises only with data + validation)

Validation Gauntlet

Why most backtests fail live — and how we refuse to

The 5 ways backtests lie (and our defence)

Data leakage / look-ahead

signal at close(t) -> trade at open(t+1); HTF & macro shifted to release

Overfitting / curve-fit

default params; in-sample vs out-of-sample split always reported

Multiple testing

few strategies tested; deflated-Sharpe mindset; no p-hacking

Ignoring costs

every trade net of spread+slippage; cost-sensitivity run

One-regime sample

regime-stratified results; flagged as a known gap

The gate (in order)

Temporal firewall (embargo)

↓

Walk-forward + Purged CV

↓

Regime-stratified backtest

↓

30-day paper trading

↓

Calibration (Brier<0.25)

↓

Go / No-Go certification

BUILDING NEXT: walk-forward + Combinatorial Purged CV (López de Prado) — the rigor gate that lets us trust ANY strategy or model before capital.

Research Evidence & Model Selection

What the peer-reviewed literature actually supports (deep-research, adversarially verified)

Topic	Finding	Evidence	NexaQuant decision
Stop-loss	Regime-conditional: HELPS under momentum, HURTS under random-walk/mean-reversion *Kaminski & Lo 2014*	**STRONG**	VALIDATES us — we only stop within the TREND regime
Let winners run	Fixed/MACD take-profit did NOT beat letting it run (FX/metals/crypto) *Vezeris 2018*	**MODERATE**	VALIDATES — momentum-ride > 2R target in our tests
Stop+momentum	10% stop on momentum: Sharpe 0.17->0.37, skew flipped +; tail capped *Han-Zhou-Zhu 2016*	**STRONG***	Supports SL + momentum (but equity, gross of cost)
Vol-targeting	NAIVE single-factor vol-targeting fails OOS & net of costs *Cederburg 2020*	**STRONG(-)**	CAUTION — matches our cost-churn finding; size cost-aware
Vol (multifactor)	Multifactor, cost-optimized vol-conditioning DOES add ~13% Sharpe *DeMiguel 2024*	**STRONG**	Adopt at portfolio level, not naive per-bar
CPCV + Deflated SR	Adaptive search -> spurious backtests; needs trial-adjusted metrics *Bailey & LdP; Harvey-Liu*	**STRONG**	VALIDATES our validator.py (CPCV + DSR + embargo)
Deep RL	Crypto DRL prone to false-positive overfit; needs overfit test *Gort et al. 2022*	**MODERATE**	VALIDATES deferring RL behind the gate
SMC (FVG/OB/ICT)	ZERO verified academic evidence of independent edge *(unanswered)*	**NONE**	Treat as folklore; our edge = trend/momentum overlap
Free data	Dukascopy, CFTC COT, FRED are primary/trusted sources *Dukascopy/CFTC/FRED*	**n/a**	Our pullers target these (+ Binance, Stooq)

SCOPE CAVEAT + WHAT TO AVOID

Most STRONG evidence is monthly US EQUITY factors — NOT intraday gold/BTC. Mechanisms transfer, but we must RE-VALIDATE on our instruments (the rigor gate does this).
Refuted / avoid: specific ATR(12,6,2) params · Kelly-VIX hybrid sizing · 'vol-scaling doubles t-stat' · CVaR-variant superiority claims.

Validated Config & Results (v2)

NexaQuant
Gold + BTC Alpha

Design goal: make money at LOW RISK by trading only high-confidence setups

DESIGN PRINCIPLE

Win-rate / trade-selection first; lot size follows. Regime-aware LONG+SHORT, confluence entries (trend regime + structure + momentum), momentum-ride exit + scale-out, CONFIDENCE-TIERED risk (0.5% / 1% / 2% by trend strength, 2% hard cap). AI meta-label tightens selection as data grows.

Re-test — best pair & timeframe (OOS, net of cost)

pair	TF	win%	Sharpe	maxDD	verdict
BTC	H1	50%	3.66	2.5%	low-risk pick
XAU	H1	44%	4.59	3.9%	low-risk pick
BTC	H4	43%	1.95	14.8%	more return, more DD
ETH	H4	46%	-0.01	28%	weak
BNB	H1	26%	0.40	5.3%	weak
SOL	H4	42%	-2.41	33%	avoid

REALISTIC TARGET

Low risk, high Sharpe: BTC/XAU H1, long+short.
~3% max drawdown, win ~50% with big payoff.
Return ~10-25%/yr (more with capital + AI selection).
Beats a bank 3-5x at low drawdown.

HONEST LIMITS

5-10%/MONTH is NOT achievable sustainably (Medallion, the best ever, ~4%/mo). Chasing it = ruin. Bigger money = robust edge x CAPITAL x time, NOT leverage. Backtest, not live — paper-test first.

Roadmap & KPIs

The only safe path to live capital

Phases

NOW	P1	P2	P3	P4	P5
Edge proven on gold; SMC+regime+AI framework built	Pull M5/M15 + BTC; pull free fundamentals	Multi-regime history; ML ensemble	Walk-forward + Purged CV (100s of trades)	30-day paper; backtest ≈ live	Live ramp 25→50→100%; add FX
DONE	DATA	FEATURES	VALIDATE	GATE	SCALE

Live targets (realistic, post-cost)

50-60%	1.0-1.5	1-4%	< 20%	90-95%
Win rate	Sharpe	Monthly ROI	Max drawdown	Uptime

SUCCESS METRIC

A strategy that holds Sharpe ≥ 1.0 ACROSS regimes AND in 30-day paper trading earns real capital — nothing else does.
Model complexity (ML→ensemble→RL→multi-agent) increases ONLY together with data breadth and validation rigor.